Compton scattering

Thomson scattering: <u>classical description</u> of light scattering off free electrons (charges)

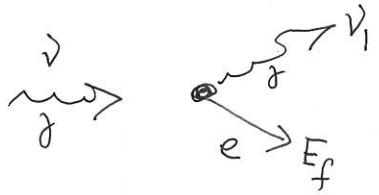
It is elastic $k_{in} = k_{out}$. $\Rightarrow \nu_{in} = \nu_{out}$

$$\frac{d\sigma_T}{d\Omega} = \frac{1}{2} r_0^2 (1+\cos^2\theta) \quad , \cos\theta = \hat{k}_{in} \cdot \hat{k}_{out} \text{ scattering angle}$$

$$r_0 = \frac{4\pi\varepsilon_0 e}{mc^2}$$

Quantum description: Compton scattering

In electron rest-frame



4-momenta

$$\vec{P}_{\gamma i} = \frac{h\nu}{c}(1, \hat{k}_i) \qquad \vec{P}_{\gamma f} = \frac{h\nu_1}{c}(1, \hat{k}_f)$$

$$\vec{P}_{ei} = mc(1,0) \quad \Longrightarrow \quad \vec{P}_{ef} = \left(\frac{E}{c}, \vec{p}\right)$$

$$\vec{P}_{\gamma i} + \vec{P}_{ei} = \vec{P}_{\gamma f} + \vec{P}_{ef}$$

$$\vec{P}_{ef} = \vec{P}_{\gamma i} + \vec{P}_{ei} - \vec{P}_{\gamma f} \Rightarrow \vec{P}_{ef}^{\,2} = \left(\vec{P}_{\gamma i} + \vec{P}_{ei} - \vec{P}_{\gamma f}\right)^2$$

But $\vec{P}_{ef}^{\,2} = \left(\frac{E}{c}\right)^2 - \vec{p}^{\,2} = \frac{m^2c^4 + c^2\vec{p}^{\,2}}{c^2} - \vec{p}^{\,2} = m^2c^2.$

$$\vec{P}_{\gamma i}^{\,2} = \vec{P}_{\gamma f}^{\,2} = 0 \quad \text{(massless!)}$$

$$\vec{P}_{ef}^{\,2} = m^2c^2 = \underset{0}{\vec{P}_{\gamma i}^{\,2}} + \underset{m^2c^2}{\vec{P}_{ei}^{\,2}} + \underset{0}{\vec{P}_{\gamma f}} + 2\underbrace{\vec{P}_{\gamma i}\cdot\vec{P}_{ei}}_{h\nu m} - 2\vec{P}_{\gamma f}\left(\vec{P}_{\gamma i} + \vec{P}_{ei}\right)$$

$$\Rightarrow \cancel{2}mh\nu = \vec{P}_{\gamma f}\cdot\vec{P}_{\gamma i} + \vec{P}_{\gamma f}\cdot\vec{P}_{ei} = \frac{h\nu}{c}\cdot\frac{h\nu_1}{c}\left(1 - \hat{k}_i\cdot\hat{k}_f\right) + \frac{h\nu_1}{c}\cdot mc$$

$$\Rightarrow h\nu = \frac{h\nu\cdot h\nu_1}{mc^2}(1-\cos\theta) + h\nu_1$$

$$h\nu_1 = \frac{h\nu}{1 + \frac{h\nu}{mc^2}(1-\cos\theta)} \Rightarrow h\nu_1 \approx h\nu\left[1 - \frac{\lambda_c \nu}{c}(1-\cos\theta)\right]$$

$$\lambda_c = \frac{h}{m_e c} = \text{Compton wavelength}$$

0.024 Å for e.

In terms of wavelength $\lambda = \frac{c}{\nu}$:

$$\frac{1 + \frac{\lambda_c \nu}{c}(1-\cos\theta)}{h\nu} = \frac{1}{h\nu_1}$$

$$\lambda \cancel{h\nu}\left[1 + \frac{\lambda_c}{\lambda}(1-\cos\theta)\right] = \cancel{h\nu}\,\lambda_1$$

$$\Rightarrow \lambda_1 - \lambda = \lambda_c(1-\cos\theta)$$

Since $1-\cos\theta \geq 0 \Rightarrow \lambda_1 \geq \lambda$ [Note that for $\lambda_c = 0$ $\lambda_1 = \lambda$ which is just classical result]

$\Rightarrow h\nu_1 \ll h\nu$

(in the electron's rest frame)

Another way: $\frac{h\nu}{mc^2} = \frac{\lambda_c}{\lambda} \ll 1$ for $h\nu \ll mc^2$.

which again corresponds to an elastic collision.

Photon scattering off moving electrons

$$\vec{P}_{\gamma i} = \frac{h\nu}{c}(1, \hat{k}_i) \qquad \vec{P}_{\gamma f} = \frac{h\nu_1}{c}(1, \hat{k}_f)$$

$$\vec{P}_{ei} = \left(\frac{E_i}{c}, \vec{p}_i\right) \Rightarrow \vec{P}_{ef} = \left(\frac{E_f}{c}, \vec{p}_f\right)$$

$$\vec{P}_{\gamma i} + \vec{P}_{ei} = \vec{P}_{\gamma f} + \vec{P}_{ef}$$

From $\vec{P}_{ef}^{\,2} = m^2c^2 = \left(\vec{P}_{\gamma f} + \vec{P}_{ef} - \vec{P}_{ei}\right)^2$

we again obtain

$$\vec{P}_{\gamma i} \cdot \vec{P}_{\gamma f} + \vec{P}_{ei} \cdot \vec{P}_{\gamma f} = \vec{P}_{\gamma i} \cdot \vec{P}_{ei}$$

$$\frac{h\nu}{c}\frac{h\nu_1}{c}\left(1 - \hat{k}_i \cdot \hat{k}_f\right) + \frac{h\nu_1}{c}\left(\frac{E_i}{c} - \vec{p}_i \cdot \hat{k}_f\right) = \frac{h\nu}{c}\left(\frac{E_i}{c} - \vec{p}_i \cdot \hat{k}_i\right)$$

For electron with initial velocity v : $E_i = \gamma mc^2$

$$\vec{k}_i \cdot \vec{k}_f = \cos\widehat{\Theta} \; ; \; \vec{p}_i \cdot \hat{k}_i = \gamma m v \cos\theta$$

$$\vec{p}_i \cdot \hat{k}_f = \gamma m v \cos\theta_1$$

$$\Rightarrow \frac{h\nu_1}{c}\left[\frac{\gamma mc^2}{c} - \gamma m v \cos\theta_1 + \frac{h\nu}{c}\left(1 - \cos\widehat{\Theta}\right)\right] =$$

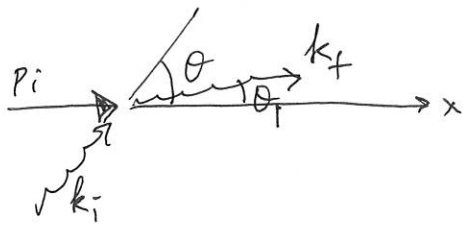
$$= \frac{h\nu}{c}\left(\frac{\gamma mc^2}{c} - \gamma m v \cos\theta\right)$$

$$\frac{h\nu_1}{h\nu} = \frac{\gamma mc(1 - \beta\cos\theta)}{\gamma mc\left(1 - \beta\cos\theta_1 + \frac{h\nu}{\gamma mc^2}\left(1 - \cos\widehat{\Theta}\right)\right)}$$

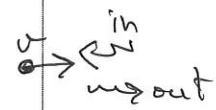
$$h\nu_1 = h\nu \frac{1 - \beta\cos\theta}{1 - \beta\cos\theta_1 + \frac{h\nu}{\gamma mc^2}\left(1 - \cos\widehat{\Theta}\right)}$$

Consider again $\frac{h\nu}{\gamma mc^2} \ll 1$

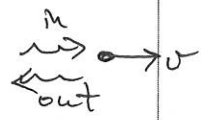
$$\Rightarrow \quad h\nu_1 \approx h\nu \, \frac{1-\beta\cos\theta}{1-\beta\cos\theta_1}$$



<u>Relativistic</u> $\theta = \pi, \ \theta_1 = 0$: $\quad \frac{\nu_1}{\nu} = \frac{1+\beta}{1-\beta} \approx \frac{2}{\frac{1}{2\gamma^2}} = 4\gamma^2 \gg 1$



$\theta = 0, \ \theta_1 = \pi$: $\quad \frac{\nu_1}{\nu} = \frac{1-\beta}{1+\beta} \approx \frac{1}{4\gamma^2} \ll 1$



$\theta = \frac{\pi}{2}, \ \theta_1 = 0$: $\quad \frac{\nu_1}{\nu} = \frac{1}{1-\beta} \approx 2\gamma^2$

<u>Non-relativistic</u> $\quad \frac{\nu_1}{\nu} \approx (1-\beta\cos\theta)(1+\beta\cos\theta_1) \approx 1 + \beta(\cos\theta_1 - \cos\theta)$

$$\Rightarrow \quad \frac{\Delta\nu}{\nu} \approx \beta \ll 1.$$